MDHGFN: Multiscale Dual Hypergraph Fusion Spatiotemporal Network for Traffic Flow Prediction

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Abstract

Traffic flow prediction is a fundamental challenge in the development of intelligent transportation systems. However, existing spatiotemporal prediction frameworks typically rely on either predefined low-order graphs or single-scale hypergraphs to model urban spatial relationships, which often fail to effectively capture the ubiquitous multiscale high-order spatial characteristics of traffic networks, thereby limiting the model's prediction accuracy and generalization capability. To address this, we propose a high-order spatial-aware traffic flow prediction framework—the Multiscale Dual Hypergraph Fusion Spatiotemporal Network (MDHGFN). Specifically, we first introduce a multiscale dual hypergraph construction method, which systematically models high-order spatial features of traffic across three scales: microscopic individual travel intent, mesoscopic community commuting patterns, and macroscopic regional flow propagation. Building upon this foundation, we design a comprehensive prediction framework integrating both spatial-aware and temporal-aware blocks. The spatial-aware block employs hypergraph convolution and dynamic graph convolution to extract dual hypergraph features progressively, incorporates a gating mechanism to fuse high-order local spatial information from multiscale dual hypergraphs, and further utilizes spatial multi-head attention to capture global spatial dependencies. The temporal-aware block effectively extracts both short-term fluctuations and long-term trends in traffic flows through dynamic temporal dilated causal convolution and temporal multi-head attention. Finally, the output layer aggregates spatiotemporal features from stacked spatiotemporal-aware blocks to generate accurate future traffic flow predictions. Extensive comparative experiments on four publicly available real-world traffic flow datasets demonstrate that the proposed model significantly outperforms 14 state-of-the-art baseline models across multiple evaluation metrics.

Keywords: Traffic flow prediction, Spatiotemporal model, Time series analysis

1. Introduction

With the rapid development of global urbanization, the surge in the number of motor vehicles has led to a series of urban transportation issues, including road congestion, frequent accidents, and environmental pollution [1, 2]. To address these challenges, Intelligent Transportation Systems (ITS) have emerged, leveraging advanced digital technologies to optimize traffic flow and improve operational efficiency [3, 4, 5]. As a core technology of ITS, traffic flow prediction provides crucial data support for traffic management, which facilitates signal optimization and flow allocation [6, 7, 8]. However, achieving high-accuracy traffic flow prediction hinges on accurately modeling the complex spatial relationships of modern cities and the temporal trends of traffic flows, which has become a focus of academic research.

Traffic flow prediction techniques have evolved from traditional statistical methods to modern deep learning approaches [9, 10]. Early studies primarily relied on traditional statistical models, such as regression analysis and time series models, to predict dynamic changes in traffic flow [11, 12, 13]. However, these methods exhibited significant limitations in capturing the spatiotemporal dependencies and nonlinear relationships inherent in traffic flow [14, 15, 16]. With the rise of machine learning, algorithms such as support vector machines [17] and random forests [18] were introduced, offering some capability to handle the nonlinear characteristics of traffic flow. However, they still struggled to model spatial dependencies effectively. Subsequently, deep learning techniques drove significant progress in traffic flow prediction. Models such as convolutional neural networks (CNN)[19] and recurrent

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neural networks (RNN)[20] excelled in capturing temporal dependencies but showed limitations in addressing the spatial dependencies of traffic flow. The introduction of graph convolutional networks (GCNs) marked a breakthrough in addressing these challenges. By learning from graph structures, researchers were able to effectively capture spatial dependencies between regions in traffic networks [21]. Building on this foundation, several advanced models have been proposed in recent years for high-accuracy prediction in complex traffic scenarios. For instance, DHSTNet [22] employs a dynamic spatiotemporal graph model that integrates spatiotemporal dependencies with dynamic graph structures, providing a more comprehensive representation of traffic dynamics. Similarly, STGHTN [23] introduces a spatiotemporal gated hybrid transformer network, which combines local features from temporal gated convolutions, spatial features from gated graph convolutions, and global features from transformers, further improving prediction accuracy.

Although significant progress has been made in traffic flow prediction research, current methods still exhibit notable limitations in spatial feature modeling. The prevailing spatiotemporal prediction frameworks primarily adopt two modeling paradigms: node-edge representation based on predefined low-order graph structures, or hyperedge feature learning relying on single-scale hypergraphs. While these approaches can capture certain spatiotemporal dependencies, they fail to adequately characterize the ubiquitous multiscale higher-order spatial features inherent in traffic networks, consequently limiting their predictive performance in complex urban traffic scenarios [24, 25, 26, 27].

In reality, the dynamic evolution of traffic flows is shaped by the combined effects of multiscale factors, including microscopic travel intentions, mesoscopic community commuting patterns, and macroscopic regional flow propagation, respectively. At the microscopic level, heterogeneous travel demands (e.g., school trips vs. medical visits) generate superimposed complex patterns within the same spatiotemporal region, leading to abrupt fluctuations in traffic volume [28, 29, 30, 31]. The mesoscopic scale reveals spatial mismatches between administrative boundaries and actual commuting corridors (e.g., educational districts extending beyond jurisdictional divisions), resulting in clustered traffic flow distributions that transcend administrative units [32, 33]. Furthermore, macroscopic propagation effects demonstrate how non-adjacent traffic networks amplify incident impacts through detour routing, triggering cascading flow variations across spatially discontinuous regions [14, 34, 35]. Consequently, comprehensive integration of these multiscale factors proves essential for holistically capturing spatiotemporal dependencies in traffic flows, while simultaneously offering novel research perspectives for enhancing prediction accuracy.

To address this, we propose a traffic flow prediction framework with higher-order spatial awareness—the Multiscale Dual Hypergraph Fusion Spatiotemporal Network (MDHGFN). Specifically, we first introduce a multiscale dual hypergraph construction method, which systematically models high-order spatial features of urban traffic across three scales: microscopic individual travel intent, mesoscopic community commuting patterns, and macroscopic regional flow propagation. Building upon this foundation, we design a comprehensive prediction framework integrating both spatial-aware and temporal-aware blocks. The spatial-aware block employs hypergraph convolution and dynamic graph convolution to extract dual hypergraph features progressively, incorporates a gating mechanism to fuse high-order local spatial information from multiscale dual hypergraphs, and further utilizes spatial multi-head attention to capture global spatial dependencies. The temporal-aware block effectively extracts both short-term fluctuations and long-term trends in traffic flows through dynamic temporal dilated causal convolution and temporal multi-head attention. Finally, the output layer aggregates spatiotemporal features from stacked spatiotemporal-aware blocks to generate accurate future traffic flow predictions.

The contributions of this study are summarized as follows:

- We propose a traffic flow prediction model with multiscale higher-order spatial awareness, which dynamically captures multiscale higher-order spatial features and complex temporal dependencies to achieve accurate traffic flow forecasting.
- We introduce a multiscale dual hypergraph construction method, which systematically characterizes higher-order spatial features of traffic networks from three perspectives: microscopic individual travel intent, mesoscopic community commuting patterns, and macroscopic regional flow propagation, respectively.
- We design a spatiotemporal perception method based on multiscale hypergraph feature fusion, adaptively integrating higher-order spatial features at the microscopic, mesoscopic, and macro-scales, effectively enhancing the model's ability to capture multiscale dynamic interaction patterns.
- Extensive experiments on four real-world traffic flow datasets demonstrate that the proposed model significantly outperforms 14 state-of-the-art baseline models across multiple evaluation metrics, validating the superiority and generalization ability of the proposed approach.

2. Related Work

2.1. Graph and Hypergraph

The graph represents real-world network structures, such as social networks, through relationships between nodes and edges, and has been widely applied across various fields. In the context of intelligent transportation, graph structures are extensively used to describe the relationships between intersections or road segments within urban

traffic networks. Based on this, researchers have introduced graph convolution networks, which combine graph structures with convolution operations to extract spatial relationships in urban traffic networks and predict traffic flow dynamics. For instance, GCRN [21] models the spatial dependencies of traffic networks through graph structures and employs a data-driven approach to optimize the graph structure, thereby improving the accuracy of traffic flow prediction.

Compared to traditional graph structures, hypergraphs can capture more complex multi-node relationships, offering significant advantages in revealing higher-order correlations. With their ability to model complex interactions in traffic flow, hypergraphs have gradually been applied in the field of traffic flow prediction. For instance, MGHCN [36] transforms traditional graph edges into hypergraph nodes using a dual-hypergraph structure, effectively capturing higher-order correlations between edges. Similarly, STGHCN [37] leverages hypergraph structures to capture both pairwise and higher-order spatial patterns among road nodes while exploring the correlations between different types of traffic data, significantly improving traffic flow prediction accuracy. However, existing hypergraph-based traffic flow prediction models generally suffer from unidimensional modeling limitations, failing to effectively characterize and integrate the intrinsic multiscale spatial features (e.g., microscopic travel behaviors, mesoscopic regional interactions, and macroscopic network structures) inherent in urban transportation systems, which significantly constrains their prediction performance and generalization capability in complex urban traffic scenarios.

2.2. Graph and Hypergraph Convolution Networks

Graph convolution networks (GCNs), initially proposed by Kipf et al., leverage information from neighboring nodes to update node features [38]. In each GCN layer, nodes aggregate the features of their direct neighbors through an aggregation function, which is then integrated to update their feature representations. This mechanism enables GCNs to effectively capture the spatial structure of urban traffic networks and has been widely applied in traffic flow prediction. For instance, A3T-GCN [39] employs GCNs to learn spatial dependencies based on the topology of traffic networks, while ST-GCN [40] extracts spatiotemporal features from graph-structured time series data using graph convolution operations.

Compared to GCNs, hypergraph convolution networks (HGCNs) perform convolution operations on hypergraphs, enabling a more comprehensive extraction of higher-order relationships and complex interaction patterns between nodes. This enhanced modeling capability has drawn increasing attention to HGCNs in the field of traffic flow prediction. For instance, DDSTGCN [9] transforms traffic flow graphs into dual hypergraphs to capture the dynamic spatiotemporal features of edges. By performing collaborative convolutions on both the traffic flow graph and its dual hypergraph, DDSTGCN effectively improves traffic flow prediction performance.

2.3. Dynamic Convolution Networks

Dynamic graph convolutional networks (DGCNs) extract node features through convolution operations while capturing the dynamic evolution of graph structures over time [41]. This capability enables DGCNs to effectively model the dynamic evolution characteristics of transportation networks, making them widely applied in traffic flow prediction. For instance, DGCRN [24] proposes a traffic prediction framework that combines dynamic graph convolution and recurrent neural networks. By designing a hypergraph to extract dynamic features and generate dynamic graphs, DGCRN significantly enhances the performance and efficiency of traffic flow prediction.

In recent years, DGCNs have increasingly been combined with dynamic temporal convolution networks (DTCNs) for traffic flow prediction. These models not only capture the spatial structure of spatiotemporal data but also handle the temporal dependencies that evolve over time. For instance, the GTCN model [25] integrates dynamic graph convolution with dynamic temporal convolution to capture the dynamic changes in spatiotemporal data and applies variations in the Laplacian matrix to traffic flow prediction. Similarly, Dynamic-GRCNN [42] proposes a framework based on spatiotemporal correlation dynamic graph neural networks. Modeling and learning dynamic spatiotemporal features from short-term, medium-term, and long-term historical data effectively improves the accuracy of traffic flow prediction.

2.4. Multi-head Attention

The attention mechanism effectively addresses the limitations of traditional sequential models in capturing long-term dependencies [43]. By calculating weights between elements in a sequence, it dynamically focuses on critical information in the input, enhancing the ability to capture long-term dependencies [44]. However, the limitation of a single attention head makes it challenging to comprehensively capture information across different subspaces.

To address this issue, multi-head attention is introduced to learn diverse representations of the input data. This allows the model to focus on different features and correlations simultaneously, significantly enhancing the diversity and accuracy of information capture [45]. A prominent application of multi-head attention is the transformer model, which leverages its strong parallel computing capability and ability to model long-range dependencies, achieving groundbreaking advancements in fields such as natural language processing and computer vision [46, 47]. In recent years, the transformer model has also demonstrated outstanding performance in traffic flow prediction. For instance, the Traffic Transformer [48] was the first to apply the transformer architecture to traffic flow prediction, pioneering research in this area. STGHTN [23] further

improves prediction accuracy by combining local features from temporal gated convolutions, spatial features from gated graph convolutions, and global features from the Transformer.

3. Methodology

3.1. Problem Definition

Traffic flow prediction aims to utilize historical traffic data to forecast future traffic conditions. The road network topology is modeled as a weighted directed graph $G = (V, \mathcal{E}, A)$, where the node set $V = \{v_1, v_2, \ldots, v_N\}$ represents N road segments (i.e., $|V| = N$). The edge set $\mathcal{E} = \{e_1, e_2, \ldots, e_E\}$ describes E road connections (i.e., $|\mathcal{E}| = E$), and the adjacency matrix $A \in \mathbb{R}^{N \times N}$ encodes the spatial distance weights between road segments. The traffic feature matrix $X \in \mathbb{R}^{T \times N \times d}$ contains T time steps of traffic data, where d represents the dimensionality of traffic features (e.g., speed, flow) at each road node. This study constructs a mapping function $\mathcal{F}(\cdot)$ to predict P future time steps of traffic flow $X^{(t+1):(t+P)}$ based on the past T time steps of historical traffic flow data $X^{(t-T+1):t}$ and the road network topology G. Specifically, this prediction task can be formally described as:

$$\mathcal{F}\left(X^{(t-T+1):t}, G\right) = X^{(t+1):(t+P)}. \tag{1}$$

3.2. Overall Architecture

The Multiscale Dual Hypergraph Fusion Spatiotemporal Network (MDHGFN) consists of three main components: the multiscale dual hypergraph construction module, the spatial-aware block, and the temporal-aware block. The spatial-aware block includes the Higher-order Spatial Aware Module (HSAM) and the Spatial Multi-head Attention (SMA). The temporal-aware block is composed of the Temporal-aware Enhanced Module (TAEM) and the Temporal Multi-head Attention (TMA). Figure 1 illustrates the overall architecture of MDHGFN.

3.3. Multiscale Dual Hypergraph Construction

Traditional urban spatial structure perception methods typically model spatial relationships based on predefined low-order graph structures, which struggle to effectively capture the prevalent multi-scale high-order spatial features in road networks. To address this limitation, we propose a multi-scale dual hypergraph modeling approach that constructs dual hypergraphs from three perspectives: micro-level individual travel intentions, meso-level community commuting flows, and macro-level regional traffic propagation. Specifically, the travel association dual hypergraph G^t is constructed through individual travel intention analysis to capture micro-scale spatial correlations; the functional community dual hypergraph G^c is built based on community division to represent meso-scale functional synergies; and the regional conduction dual hypergraph G^r is formulated through road traffic propagation patterns to characterize macro-scale large-range spatial interactions. Figure 2 illustrates the transformation process from low-order traffic road networks to multi-scale dual hypergraphs.

Travel Intent Dual Hypergraph Existing traffic flow prediction studies commonly adopt homogeneous scenario assumptions, such as assuming that all commuters concentrate on traveling to central business districts during peak hours. Such assumptions forcibly compress diverse travel intentions into a single commuting purpose, while in reality, traffic systems exhibit multiple types of travel demands (e.g., school drop-offs, medical visits, freight transportation) even within the same time period, with significant differences in route choices and time arrangements among different groups. Consequently, existing models struggle to accurately capture the spatio-temporal distribution characteristics of traffic flow caused by the diversity of individual travel intentions.

To better understand the impact of micro-level individual diverse travel intentions on traffic flow, we propose the travel association dual hypergraph. Specifically, as shown in Figure 2(a), all nodes originating from the same road node and all nodes destined for the same road node are respectively grouped into hyperedges. The travel association dual hypergraph is represented as $G^t = (V^t, \mathcal{E}^t, H^t)$, with its adjacency matrix H^t defined as:

$$H^t_{ij} = \begin{cases} 1, & \text{if} \quad v^t_i \in e^t_j \\ 0, & \text{otherwise} \end{cases}, \tag{2}$$

where $v^t_i \in V^t$ represents road nodes and $e^t_j \in \mathcal{E}^t$ represents hyperedges. The travel association dual hypergraph H^t can be divided into two parts: $H^t = H^t_{src} + H^t_{dst}$, where H^t_{src} represents the hyperedge connection relationships based on origins, and H^t_{dst} represents those based on destinations. Together, they constitute the complete topological structure of the travel association dual hypergraph.

Functional community Dual Hypergraph

Existing traffic flow prediction studies generally overlook the high-order spatial correlation characteristics at the meso-level functional community scale when modeling urban spatial relationships. However, in real-world urban systems, communities are not strictly defined by administrative boundaries but rather emerge as dynamic functional units formed by spatio-temporal behavioral patterns such as high-frequency commuting corridors and periodic school-run flows. For instance, regular commuting corridors form between educational districts and surrounding residential areas, and these high-order spatial interaction clusters based on commuting demands essentially define the functional boundaries of communities.

To capture the impact of such dynamic functional units' spatial correlation characteristics on traffic flow, we propose the functional community dual hypergraph. Specifically, as shown in Figure 2(b), the construction process

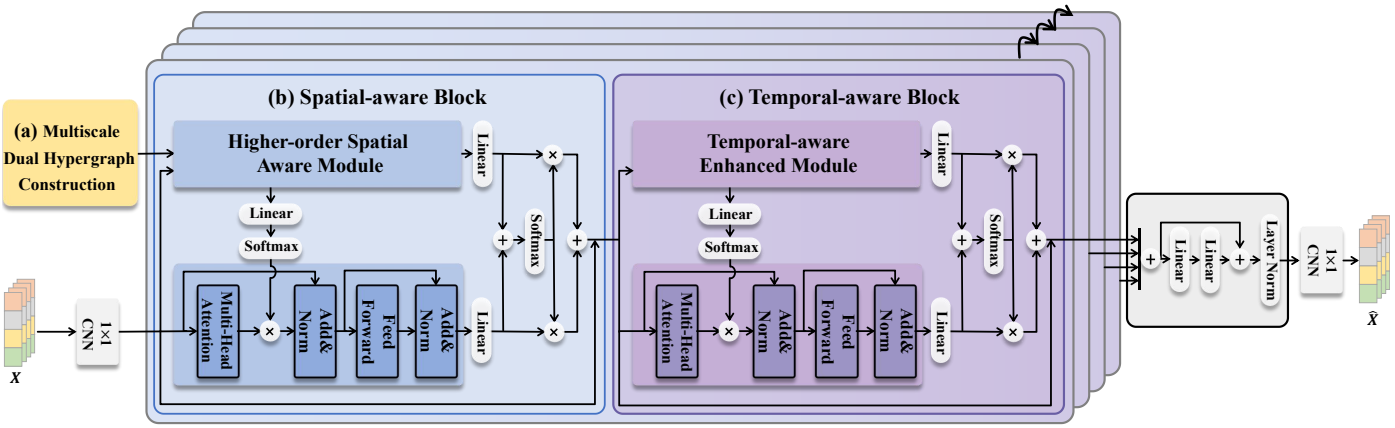


Figure 1: The proposed Multiscale Dual Hypergraph Fusion Spatiotemporal Network (MDHGFN) consists of three core components: (a) a multiscale dual hypergraph construction module that transforms the original traffic network into hypergraph representations through three distinct scales - microscopic (individual travel intent), mesoscopic (community commuting patterns), and macroscopic (regional flow propagation); (b) a spatial-aware block comprising a Higher-order Spatial-Aware Module (HSAM) that integrates hypergraph convolution and dynamic graph convolution, along with Spatial Multi-head Attention (SMA), designed to extract and fuse higher-order spatial features from local to global scales; (c) a temporal-aware block incorporating a Temporal-Aware Enhanced Module (TAEM) with dilated causal convolution and Temporal Multi-head Attention (TMA) for capturing both short-term fluctuations and long-term trends in traffic flow. The output layer synthesizes the multiscale spatiotemporal features through linear transformation and residual connections to achieve accurate traffic flow prediction.

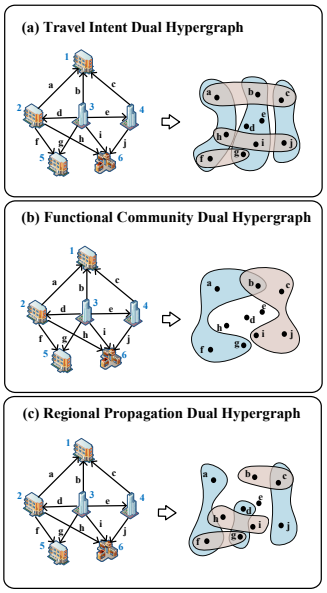


Figure 2: Schematic of the multiscale dual hypergraph construction method: (a) Travel intent dual hypergraph (G^t) at microscopic scale, where hyperedges dynamically aggregate nodes based on individual travel intent; (b) Functional community dual hypergraph (G^c) at mesoscopic scale, constructing hyperedges through community clustering of commuting behavior patterns; (c) Regional propagation dual hypergraph (G^r) at macroscopic scale, representing cross-regional flow transmission paths via hyperedges. The left column displays the original node-edge network structure (nodes 1–6), while the right column shows the transformed dual hypergraph representation with hyperedges denoted by irregular closed curves.

begins by selecting the K road nodes with highest destination coincidence to form destination hyperedges, and then selecting the K road nodes with highest origin coincidence to form origin hyperedges. During node selection, the total spatial distance within each hyperedge is constrained to be minimized, thereby constructing the dual hyperedges of the functional community hypergraph.

To quantify the functional similarity between nodes, a new functional association matrix A^c is reconstructed from the original weight adjacency matrix A, calculated as:

$$A^c_{i,j} = \sqrt{\sum_{d=1}^{N}(A_{i,d} - A_{j,d})^2} \tag{3}$$

where N is the number of nodes, and smaller values of $A^c_{i,j}$ indicate higher functional similarity between nodes v_i and v_j. Based on this, for each node i, the K-nearest neighbors with highest functional similarity are selected from A^c to form the functional community $\mathcal{E}^c_i$, ultimately constructing the dual hypergraph $G^c = (V^c, \mathcal{E}^c, H^c)$, with adjacency matrix H^c defined as:

$$H^c_{ij} = \begin{cases} 1, & \text{if} \quad v^c_i \in e^c_j \\ 0, & \text{otherwise} \end{cases}, \tag{4}$$

where road node $v^c_i \in V^c$ and hyperedge $e^c_i \in \mathcal{E}^c$. The adjacency matrix H^c of the functional community dual hypergraph can be divided into two parts: $H^c = H^c_{src} + H^c_{dst}$. Here, H^c_{src} represents all hyperedge connection relationships constructed based on origin node coincidence, while H^c_{dst} represents those based on destination node coincidence. Together, they constitute the complete topological structure of the functional community dual hypergraph.

Regional propagation dual hypergraph Traditional urban spatial modeling methods often overlook the macroscopic influence mechanisms of cross-regional traffic flow conduction. For example, in urban traffic, even if there is no direct road connection between two regions, when an

accident occurs in one region, drivers may choose alternative routes, thereby indirectly affecting the traffic flow in another region. This dynamic macroscopic conduction effect is difficult to capture with static adjacency matrices.

To address this challenge, we propose constructing a regional conduction dual hypergraph to learn the macroscopic influence relationships between regions. Specifically, as shown in Figure 2(c), the method first employs a Top-K sampling approach to select the top K most influential neighbor nodes for constructing the node influence adjacency matrix A^r, calculated as follows:

$$\begin{aligned} idx &= \text{Top-K}(-A_{i,:}, K), \\ idy &= \text{Top-K}(-A_{:,j}, K), \\ A^r &= A[idx, idy], \end{aligned} \tag{5}$$

where idx and $idy \in \mathbb{R}^{N\times K}$ respectively represent the top K most influential neighbor nodes for each node, selected based on outgoing (from the start node) and incoming (to the end node) influences.

Subsequently, the $top-K$ influential neighbor sets for each node are used as hyperedges, and based on the node influence matrix A^r, the regional conduction dual hypergraph $G^r = (V^r, \mathcal{E}^r, H^r)$ is constructed, with its adjacency matrix H^r defined as:

$$H^r_{ij} = \begin{cases} 1, & \text{if} \quad v^r_i \in e^r_j \\ 0, & \text{otherwise} \end{cases}, \tag{6}$$

where road node $v^r_i \in V^r$ and hyperedge $e^r_j \in \mathcal{E}^r$. The adjacency matrix H^r of the regional conduction dual hypergraph can be divided into two parts: $H^r = H^r_{src} + H^r_{dst}$, where H^r_{src} and H^r_{dst} respectively correspond to the hyperedge relationships based on origins and destinations, together forming the complete topological structure of the regional conduction dual hypergraph.

3.4. Spatial-aware Block

To better capture the complex and dynamic spatial relationships in traffic data, we propose a Higher-order Spatial Aware Module (HSAM) for learning local higher-order spatial embeddings, as illustrated in Figure 3. However, traffic flow in traffic networks is often influenced by traffic conditions in distant regions. Therefore, we further introduce Spatial Multi-head Attention (SMA) to capture global spatial dependencies. Finally, through interaction and fusion, the extracted local higher-order spatial embeddings are combined with global spatial dependencies to obtain the final spatial-aware features.

3.4.1. Higher-order Spatial Aware Module

To capture the high-order dynamic features of traffic flow data, a multi-scale hyperedge feature matrix $\mathbf{X}_h$ is first constructed, integrating the original traffic flow features X into the multi-scale dual hypergraph. This process is formally expressed as:

$$\mathbf{X}_h = \left[(W_1X)\left[\ \text{ind}_{src}, :\right]; (W_2X)\left[\text{ind}_{dst}, :\right]\right] \in \mathbb{R}^{E\times 2d'}, \tag{7}$$

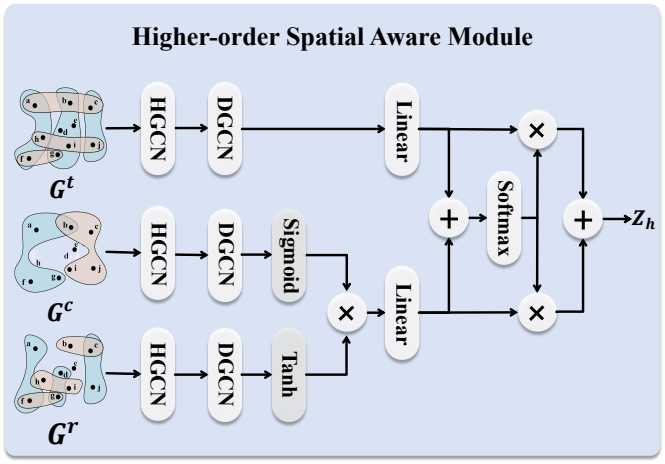


Figure 3: Schematic representation of the higher-order spatial aware module (HSAM). The module processes three dual hypergraph inputs (G^t, G^c, G^r) through parallel Hypergraph Convolutional Networks (HGCN) and Dynamic Graph Convolutional Networks (DGCN) layers to extract multiscale higher-order local spatial features. The extracted features undergo linear transformations with subsequent activation functions (Sigmoid and Tanh), followed by feature fusion to generate the comprehensive higher-order local spatial representation Z_h.

where $[\cdot;\cdot]$ denotes the concatenation operation, $W_1, W_2 \in \mathbb{R}^{T\times d\times d'}$ are learnable parameters for linearly transforming the original traffic flow features X, resulting in $W_1X, W_2X \in \mathbb{R}^{N\times d'}$. Here, $(W_1X)[\text{ind}_{src}, :]$ represents the hyperedge feature slice tensor constructed based on origin hyperedge relationships, and $(W_2X)[\text{ind}_{dst}, :]$ represents that based on destination hyperedge relationships. Next, convolution operations are applied to further aggregate interactions between super-nodes:

$$\mathbf{X}_h{}' = \text{Conv}_{1\times 1}(\mathbf{X}_h) \in \mathbb{R}^{E\times d''}, \tag{8}$$

where $\text{Conv}_{1\times 1}(\cdot)$ denotes convolution using 1×1 kernels to update super-node features, with d'' being the super-node feature dimension. To enhance model adaptability, diagonal adaptive learning parameters for the dual hypergraph are defined as follows:

$$W_{adp} = \text{diag}(L_{adp}) \in \mathbb{R}^{N\times N}, \tag{9}$$

where $L_{adp} \in \mathbb{R}^N$ is a weight vector for adaptive learning of hyperedges. Based on this, the adaptive adjacency matrix for the dual hypergraph is defined as:

$$W_h = {D_{hv}}^{-\frac{1}{2}} H W_{adp} {D_{he}}^{-\frac{1}{2}} H^\top {D_{hv}}^{-\frac{1}{2}} \in \mathbb{R}^{E\times E}. \tag{10}$$

where D_{hv} is the hyperedge degree diagonal matrix, D_{he} is the super-node degree diagonal matrix, and H is the dual hypergraph adjacency matrix.

The adaptive adjacency matrix W_h is then input into the hypergraph convolution network to aggregate high-order information from hyperedges, computed as:

$$HGCN_\Theta\left(\mathbf{X}_h{}'\right) = \sum_{n=0}^{N-1} (W_h)^n \mathbf{X}_h{}' \Theta_n \in \mathbb{R}^E. \tag{11}$$

Here, $\Theta = [\Theta_0, \ldots, \Theta_{N-1}] \in \mathbb{R}^{N \times d''}$ represents learnable parameters. Furthermore, the Reshape() function maps the output of the hypergraph convolution network to a sparse matrix:

$$D_f = \text{Reshape}\left(HGCN_{\Theta''_f}\left(\mathbf{X}_h{}'\right)\right) \in \mathbb{R}^{N \times N}. \tag{12}$$

Based on this, an inverse correlation matrix $D_b = D_f^\top$ is constructed, and dynamic graph convolution networks are used to aggregate high-order local information of traffic flow:

$$\begin{aligned} \text{DGCN}(X) = &\sum_{n=0}^{N-1} \left(A_f \odot D_f\right)^n X\Theta'_{n,f} \\ &+ \left(A_b \odot D_b\right)^n X\Theta'_{n,b} \\ &+ \left(A_{adp}\right)^n X\Theta'_{n,adp}, \end{aligned} \tag{13}$$

where $A_f = A/\text{rowsum}(A)$ is the row-normalized adjacency matrix, $A_b = A^\top/\text{rowsum}(A)$ is the column-normalized adjacency matrix, and $A_{adp} = \text{Softmax}\left(\text{Relu}\left(E_1 E_2^\top\right)\right) \in \mathbb{R}^{N \times N}$ is the adaptive adjacency matrix, with $E_1, E_2 \in \mathbb{R}^{N \times c}$ being learnable parameters. Θ'_f, Θ'_b, and Θ'_{adp} are also learnable parameter matrices.

Multiple dynamic graph convolution networks ($DGCN$) are employed to enable the model to comprehensively learn high-order local spatial information from the multi-scale dual hypergraph:

$$Z^t = DGCN_t(X), \qquad Z^c = DGCN_c(X), \qquad Z^r = DGCN_r(X), \tag{14}$$

where Z^t, Z^c, and Z^r represent the high-order spatial embeddings of the travel association dual hypergraph, community commuting dual hypergraph, and regional conduction dual hypergraph, respectively. To avoid overfitting during the aggregation of high-order spatial embeddings, a gating mechanism is introduced:

$$\begin{aligned} Z^{cr} &= \text{Sigmoid}\left(Z^c\right) \odot \text{Tanh}\left(Z^r\right), \\ g_h &= \text{Sigmoid}(Z^{cr} + Z^t), \\ Z_h &= g_h \odot Z^{cr} + (1 - g_h) \odot Z^t, \end{aligned} \tag{15}$$

where $\odot$ denotes element-wise multiplication, and the adaptive weight g_h is obtained through the Sigmoid activation function, ranging between 0 and 1. Z_h represents the final high-order local spatial embedding generated by the higher-order spatial awareness module.

3.4.2. Spatial Multi-head Attention

To effectively model the global spatial dependencies in road networks, the input feature $X \in \mathbb{R}^{T \times N \times d}$ is first transformed through three independent linear layers to generate the query matrix Q_S, key matrix K_S, and value matrix V_S as follows:

$$Q_S = XW_S^Q, \quad K_S = XW_S^K, \quad V_S = XW_S^V, \tag{16}$$

where W_S^Q, W_S^K, and W_S^V are learnable parameter matrices of the linear layers.

Next, spatial attention scores between nodes are computed by introducing a scaling factor $\sqrt{c_s}$ to stabilize gradient calculations and prevent saturation of the softmax function due to large dot product results. The attention score matrix is calculated as:

$$A_S(X) = \text{Softmax}\left(\frac{Q_S K_S^\top}{\sqrt{c_s}}\right) V_S. \tag{17}$$

Considering the limited expressiveness of a single attention mechanism, a multi-head attention mechanism is introduced to capture feature interactions across different subspaces. Specifically, the outputs of multiple attention heads are concatenated and transformed linearly to produce the final output:

$$\text{SMA}(X) = \text{Concat}(A_S^1(X), A_S^2(X), \ldots, A_S^m(X))W_S, \tag{18}$$

where m is the number of attention heads, and W_S is the output projection matrix.

To enhance the expressive power of local features, a feature interaction mechanism is designed to stabilize the training process through residual connections and layer normalization, while introducing non-linear transformations to strengthen feature representations. This results in the global spatial feature X_h^S, computed as:

$$\begin{aligned} \dot{X}_h^S &= \text{LayerNorm}(X + Z_h \odot \text{SMA}(X)), \\ \ddot{X}_h^S &= \text{ReLU}(\dot{X}_h^S), \\ X_h^S &= \text{LayerNorm}(\dot{X}_h^S + \ddot{X}_h^S), \end{aligned} \tag{19}$$

where ReLU is the activation function and LayerNorm is the layer normalization function.

Finally, to adaptively balance high-order local spatial embeddings and global spatial dependencies, an Entangle fusion mechanism is employed. This mechanism dynamically adjusts the contribution of both features using a gating weight, resulting in the final output feature that encompasses both local and global spatial information:

$$\begin{aligned} g_s &= \text{Sigmoid}(X_h^S + Z_h), \\ X_S &= g_s \odot X_h^S + (1 - g_s) \odot Z_h, \end{aligned} \tag{20}$$

where the adaptive weight g_s is obtained through the Sigmoid activation function, ranging between 0 and 1. The output feature $X_S \in \mathbb{R}^{N \times T \times d}$ retains detailed high-order local spatial information while incorporating global spatial dependencies, providing richer spatial feature representations for subsequent traffic flow predictions.

3.5. Temporal-aware Block

In real-world scenarios, traffic flow is frequently influenced by external factors such as unexpected accidents or temporary construction, where these short-term variations are crucial for ensuring prediction accuracy. To address this challenge, we propose a Temporal-aware Enhanced

Module (TAEM) to model short-term temporal dynamics. Furthermore, considering the periodic characteristics of traffic flow (e.g., rush hours and holiday patterns), we introduce Temporal Multi-head Attention (TMA) to capture long-term temporal dependencies. The comprehensive spatiotemporal representations are ultimately obtained through interactive fusion of both short-term and long-term temporal relationships.

3.5.1. Temporal-aware Enhanced Module

To capture the short-term temporal dependencies in traffic flow data, dilated causal convolution is employed to learn the short-term temporal patterns of road node traffic. Specifically, dilated causal convolution introduces a dilation factor into the standard convolution operation and increases the receptive field through deep stacking networks. Given the multi-scale spatial encoded feature sequence $X_S = \{x_1^t, x_2^t, ..., x_s^t\} \in \mathbb{R}^T$ obtained from the spatial awareness block and the filter $\mathcal{F} = \{f_1, f_2, ..., f_k\} \in \mathbb{R}^K$, the dilated causal convolution at time step t is computed as:

$$X_S * \mathcal{F}(t) = \sum_{k=1}^{K} \mathcal{F}(k) x_S^{t-u(k-1)} \tag{21}$$

where u is the dilation factor that controls the skip distance, regulating the model's receptive field in the temporal dimension and facilitating the learning of short-term temporal dependencies in traffic flow data.

Simultaneously, a gating mechanism is adopted to control the output of information flow. A gated activation unit is used to extract short-term features from the output of the dilated causal convolution layer in parallel, reducing the influence of external noise. This is formally expressed as:

$$Z_T = \text{Sigmoid}(\Phi_1 * X_S + a) \odot \text{Tanh}(\Phi_2 * X_S + b) \tag{22}$$

where Φ_1 and Φ_2 are independent one-dimensional dilated causal convolutions operating along the temporal dimension, a and b are learnable bias parameters, Sigmoid() is the activation function that determines the proportion of information passed to the next layer, and Tanh() is the activation function that performs nonlinear transformation. The resulting $Z_T \in \mathbb{R}^{N \times T \times d'}$ represents the final feature representation of the temporal awareness enhancement module.

3.5.2. Temporal Multi-head Attention

To capture the periodic temporal dependencies in traffic flow data, a temporal multi-head attention mechanism is employed to learn the long-term temporal patterns of road node traffic. Specifically, the input features are first transformed into the query matrix Q_T, key matrix K_T, and value matrix V_T through three independent linear transformation layers, defined as:

$$Q_T = X_S W_T^Q, \quad K_T = X_S W_T^K, \quad V_T = X_S W_T^V \tag{23}$$

where W_T^Q, W_T^K, and W_T^V are the learnable parameter matrices of the linear layers.

Based on these matrices, temporal attention scores between nodes are computed by introducing a scaling factor $\sqrt{c_t}$ to stabilize gradient calculations and prevent saturation of the softmax function due to large dot product results. The attention score matrix is calculated as:

$$A_T(X) = \text{Softmax}\left(\frac{Q_T K_T^\top}{\sqrt{c_t}}\right) V_T. \tag{24}$$

To address the limited expressiveness of a single attention mechanism, multi-head attention is introduced to capture feature interactions across different subspaces. Specifically, the outputs of multiple attention heads are concatenated and linearly transformed to produce the final output:

$$TMA(X_S) = \text{Concat}(A_T^1(X_S), A_T^2(X_S), \ldots, A_T^m(X_S))W_T, \tag{25}$$

where m is the number of attention heads, and W_T is the linear transformation matrix that maps the output of temporal multi-head attention to the desired dimensionality.

Finally, a feature interaction mechanism is designed to enhance the representation of local features, addressing the issue of insensitivity to short-term temporal variations in the temporal multi-head attention mechanism. This results in the long-term temporal dependency feature X_T^S, computed as:

$$\begin{aligned} \dot{X}_T^S &= \text{LayerNorm}(X_S + Z_T \odot TMA(X_S)), \\ \ddot{X}_T^S &= \text{ReLU}(\dot{X}_T^S), \\ X_T^S &= \text{LayerNorm}(\dot{X}_T^S + \ddot{X}_T^S), \end{aligned} \tag{26}$$

where ReLU is the activation function, and LayerNorm is the normalization function.

To adaptively balance the contributions of short-term temporal variations and long-term temporal dependencies, an Entangle fusion mechanism is employed. This mechanism dynamically adjusts the weights of short-term dynamics and long-term temporal dependencies in historical traffic flow data through a gating mechanism, thereby obtaining spatio-temporal feature representations $\hat{X_T^S}$ that incorporate both multi-scale spatial information and long-short term temporal dependencies:

$$\begin{aligned} g_t &= \text{Sigmoid}(X_T^S + Z_T), \\ \hat{X_T^S} &= g_t \odot X_T^S + (1 - g_t) \odot Z_T, \end{aligned} \tag{27}$$

where the adaptive weight g_t is obtained through the Sigmoid activation function, with values ranging between 0 and 1. This fusion strategy ensures that the final representation effectively captures both short-term fluctuations and long-term dependencies in traffic flow patterns.

3.6. Input Layer and Output Layer

Input Layer consists of a convolutional layer whose primary function is to transform the input traffic flow data

into a higher-dimensional space, thereby enhancing the network's representational capacity. This is formally defined as:

$$X = \text{Conv}(\mathbf{X}) + W_{adp}, \tag{28}$$

where $X \in \mathbb{R}^{T \times N \times d}$ represents the convolved feature representation, and W_{adp} is a learnable positional encoding.

For the constructed hierarchical stacked spatio-temporal awareness modules, the output module fuses the multi-scale spatio-temporal feature representations from each layer through a skip connection mechanism. The computation is expressed as:

$$X_{out} = \text{Layernorm}(W_a(W_b \sum_{m=0}^{M} \hat{X_T^S}^{(m)}) + \sum_{m=0}^{M} \hat{X_T^S}^{(m)}), \tag{29}$$

where W_a and W_b are trainable parameter matrices, M denotes the number of stacked spatio-temporal awareness modules, and LayerNorm() represents the layer normalization function.

Finally, a 1×1 convolutional layer maps the fused features to the prediction results:

$$\begin{aligned} \hat{X} &= \text{Conv}(X_{out}), \\ \hat{X}^{t+1:(t+P)} &= \left(\hat{X}_{t+1}, \hat{X}_{t+2}, \ldots, \hat{X}_{t+P}\right)^{\top}, \end{aligned} \tag{30}$$

where $\hat{X}^{t+1:(t+P)} \in \mathbb{R}^{N \times P}$ represents the predicted traffic flow sequence for the next P time steps.

4. Experiments

4.1. Dataset

We evaluate the performance of the proposed MDHGFN model on four widely adopted real-world traffic flow datasets: PEMS03, PEMS04, PEMS07, and PEMS08 [49]. These datasets, collected by the California Department of Transportation (Caltrans) through its Freeway Performance Measurement System (PeMS), consist exclusively of traffic flow data from highways. They are extensively used in traffic flow prediction research, serving as standard benchmarks in studies such as FC-LSTM, DCRNN, and STGCN. Detailed statistical information for these datasets is presented in Table 1.

We follow the dataset pre-processing methodology outlined in STGHTN [37]. The dataset is divided into 60% for training, 20% for testing, and 20% for validation.

4.2. Baseline Methods

To evaluate the effectiveness of the proposed method, we conducted comparative experiments with 14 baseline methods, adhering to the parameter settings recommended in their original papers to ensure reliable and fair comparisons. A brief overview of these baseline methods is as follows:

- FC-LSTM [50]: Fully connected long short-term memory (FC-LSTM) is a time-series prediction model based on fully connected structures, suitable for modeling temporal dependencies but limited in capturing complex spatial dependencies.
- DCRNN [51]: Diffusion convolution recurrent neural network (DCRNN) models spatial dependencies using a bidirectional random walk mechanism and effectively captures temporal dependencies through an encoder-decoder framework with scheduled sampling.
- STGCN [40]: Spatio-temporal graph convolution networks (STGCN) use graph structures to model traffic flow data and employ a fully convolutional architecture to efficiently capture spatiotemporal dependencies, significantly improving training efficiency.
- Graph WaveNet [52]: Graph waveNet leverages an adaptive dependency matrix to model implicit spatial dependencies and efficiently captures long-term sequential features using stacked dilated convolutions, making it suitable for complex traffic flow prediction tasks.
- STSGCN [53]: Spatial-temporal synchronous graph convolution Networks (STSGCN) capture complex local spatiotemporal correlations through synchronous spatiotemporal modeling and employ multiple modules to handle heterogeneity across different time periods, improving prediction accuracy.
- OGCRNN [54]: Optimized graph convolution recurrent neural network (OGCRNN) is an optimized GCRNN that learns a data-driven graph structure during training to better capture latent spatial relationships, combined with temporal sequence modeling to significantly improve performance.
- STGODE [55]: Spatial-temporal graph ordinary differential equation networks (STGODE) is a tensor-based differential equation model that simultaneously captures spatiotemporal dynamics and models long-range dependencies in complex traffic networks using semantic adjacency matrices and dilated temporal convolutions.
- STFGNN [56]: Spatial-temporal fusion graph neural networks (STFGNN) address missing potential associations in traditional spatial graphs by generating data-driven temporal graphs and efficiently learn hidden spatiotemporal dependencies by fusing multiple spatiotemporal graph structures combined with gated convolution modules for long-sequence feature processing.
- GMSDR [57]: Graph-based multi-step dependency relation (GMSDR) combines graph neural networks

Table 1: Statistics of the PEMS03, PEMS04, PEMS07, and PEMS08 datasets, including the number of nodes, edges, time steps, and date range.

Dateset	Nodes Number	Edges Number	Time Steps	Date Range
PEMS03	358	547	26208	09/01/2018 - 11/30/2018
PEMS04	307	340	16992	01/01/2018 - 02/28/2018
PEMS07	883	866	28224	05/01/2017 - 08/31/2017
PEMS08	170	295	17856	07/01/2016 - 08/31/2016

with multi-step dependency relations (MSDR), explicitly leveraging hidden states from multiple historical time steps embedded with spatial information to effectively capture complex long-term spatiotemporal dependencies.

- FEDformer [58]: Frequency enhanced decomposed transformer (FEDformer) combines seasonal-trend decomposition with frequency-domain enhancement techniques, improving the ability to model long-term time series while maintaining linear time complexity, making it suitable for efficient prediction tasks.

- HSTGCNT [59]: Hierarchical spatio-temporal graph convolution networks with transformer (HSTGCNT) integrates a long-term temporal transformer (LTT) module and a spatiotemporal graph convolution (STGC) module for short-term temporal and spatial relationship modeling. It fuses long- and short-term dependencies through attention mechanisms while alleviating over-smoothing issues in graph convolution networks.

- AGFCRN [60]: Adaptive graph fusion convolution recurrent network (AGFCRN) dynamically captures spatiotemporal dependencies using self-learned node embeddings to generate static and dynamic graph structures. It combines gated recurrent units and attention mechanisms to enhance long-term sequence prediction.

- DS-STGCN [14]: Dynamic multi-scale spatial-temporal graph convolution network (DS-STGCN) introduces an innovative attention layer for dynamic spatial dependency modeling and employs a multi-branch temporal network to learn multi-scale temporal features from both the time and frequency domains, adapting to complex traffic patterns.

- CDAGF [61]: Contrastive learning-based adaptive graph fusion network with residual-enhanced decomposition (CDAGF) proposes a graph-enhanced contrastive learning mechanism to address data scarcity and employs residual-enhanced decomposition to separate abnormal signals. It dynamically fuses static and dynamic graphs for spatiotemporal feature modeling, improving prediction accuracy.

4.3. Evaluation Criteria

We employ three widely adopted metrics to evaluate the predictive performance of the model, namely Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). Specifically, MAE is calculated as the average of absolute differences between predicted and ground-truth values, providing a comprehensive measure of overall prediction deviation. RMSE is computed as the square root of the mean squared differences between predictions and actual observations, exhibiting heightened sensitivity to larger errors and thereby enabling effective assessment of the model's predictive accuracy under peak traffic conditions. MAPE is determined by averaging the absolute percentage differences between predicted and true values, which highlights relative prediction errors and facilitates the evaluation of the model's performance in scenarios with low traffic flow. The formulas for these metrics are as follows:

$$\begin{aligned}
\mathrm{MAE} &= \frac{1}{N \times P} \sum_{t=1}^{P} \sum_{n=1}^{N} \left| X_t^n - \hat{X}_t^n \right|, \\
\mathrm{RMSE} &= \sqrt{\frac{1}{N \times P} \sum_{t=1}^{P} \sum_{n=1}^{N} \left(X_t^n - \hat{X}_t^n \right)^2}, \\
\mathrm{MAPE} &= \frac{1}{N \times P} \sum_{t=1}^{P} \sum_{n=1}^{N} \frac{\left| X_t^n - \hat{X}_t^n \right|}{X_t^n},
\end{aligned} \tag{31}$$

where, N represents the total number of observed samples, P denotes the number of prediction time steps, X_t^n is the ground truth value for the n-th sample at time t, and $\hat{X}_t^n$ is the predicted value for the n-th sample at time t.

4.4. Experimental Settings

All experiments in this study are conducted on a Linux operating system. The hardware environment includes an Intel(R) Xeon(R) W-2245 CPU @ 3.90GHz, 32GB of RAM, and an Nvidia 4090 GPU. During model training, the Adam optimizer is used, and hyperparameters are optimized via grid search. The batch size is set according to the characteristics of the datasets: for the PEMS04, PEMS08, and PEMS03 datasets, the batch size is set to 16, while for the PEMS07 dataset, due to hardware constraints, the batch size was set to 6. The learning rate is uniformly set to 0.001 to ensure the comparability of experimental results.

Regarding model architecture, the proposed framework consists of four stacked spatiotemporal aware blocks. Each

block integrates a multi-head attention mechanism, with the channel size set to 32 and the number of attention heads set to 4. This design aims to effectively capture spatiotemporal features, thereby enhancing the model's prediction performance.

4.5. Experimental Results

4.5.1. Performance Comparisons

Table 2 presents the experimental results of different models on the PEMS03, PEMS04, PEMS07, and PEMS08 datasets for 60-minute traffic flow prediction. Compared to other baseline methods, MDHGFN achieves significant performance improvements across all evaluation metrics. A detailed analysis and discussion of the experimental results are provided below.

Traditional methods (e.g., FC-LSTM and DCRNN) perform relatively poorly across all datasets due to their inability to effectively model spatial dependencies, indicating that relying solely on simple time series modeling is insufficient for addressing the complexity of traffic flow prediction. Graph convolution-based models (e.g., STGCN, Graph WaveNet, and STSGCN) achieve significant performance improvements over traditional methods by leveraging graph structures to capture spatial features. However, they still face limitations in modeling long-term sequence dependencies and dynamic spatial relationships. Further optimized approaches (e.g., STFGNN, GMSDR, and OGCRNN) incorporate adaptive graph structures and multi-step dependency modeling, showing superior performance on certain datasets. Nevertheless, their performance on large-scale datasets (e.g., PEMS03 and PEMS07) remains limited, as they struggle to fully capture the intricate spatiotemporal characteristics of traffic data. Transformer-based models (e.g., FEDformer and HSTGCNT) excel in long-term sequence forecasting by enhancing temporal modeling capabilities. However, they still exhibit certain limitations in effectively capturing complex spatial dependencies. In recent years, several innovative strategies have been proposed. For instance, DS-STGCN introduces attention mechanisms to dynamically model spatial dependencies and leverages a multi-branch temporal network to learn multi-scale temporal features from both the time and frequency domains. Similarly, CDAGF dynamically fuses static and dynamic graphs to model spatiotemporal characteristics. However, these methods are still limited by the expressive capacity of graph structures and struggle to fully extract higher-order information embedded in traffic flow data.

In contrast, MDHGFN leverages hypergraph convolution to integrate multiscale higher-order information, such as human behavior, community structure, and physical connectivity, into the model, enabling precise characterization of complex spatial features in urban traffic. The evaluation metrics across all datasets achieve state-of-the-art results. For instance, on the PEMS03 dataset, MDHGFN achieves an MAE of 14.12, RMSE of 22.52, and MAPE of 12.95, representing improvements of 11.9% (MAE), 14.7% (RMSE), and 10.7% (MAPE) compared to the second-best baseline model. This trend is also validated on other datasets, further demonstrating the superiority, adaptability, and robustness of MDHGFN in diverse traffic scenarios. Notably, MDHGFN performs better on large-scale datasets (e.g., PEMS03 and PEMS07) than on smaller ones (e.g., PEMS04 and PEMS08). This is because in large-scale datasets, the traffic networks contain more prominent and intricate high-order dependencies. MDHGFN's hypergraph convolution, which treats edges as hypernodes, is well-suited to capture these high-order dependencies effectively, enabling it to better uncover latent patterns and improve prediction accuracy. In contrast, smaller datasets have simpler network structures with fewer high-order dependencies, which limits the model's ability to showcase its full performance potential. Furthermore, among the two large-scale datasets, MDHGFN achieves more significant performance improvements on PEMS03 than on PEMS07. This is primarily due to PEMS03 having a substantially higher number of edges relative to nodes. This structural characteristic aligns well with the design of MDHGFN's hypergraph convolution (which treats edges as hypernodes), allowing the model to more fully capture the high-order dependencies in the complex traffic network of PEMS03, thereby achieving better prediction results.

4.5.2. Ablation Studies

To verify the effectiveness of each component in MDHGFN, ablation experiments are conducted on the PEMS03 and PEMS08 datasets, and the results are analyzed. Based on the original model, three model variants are designed by progressively removing key components to evaluate their impact on overall performance. The specific designs are as follows:

- MDHGFN w/o both HSAM and TAEM: In this variant, both the higher-order spatial aware module (HSAM) and the temporal-aware enhanced module (TAEM) are removed. This variant is designed to evaluate the model's performance when both temporal feature modeling and higher-order spatial dependency modeling capabilities are entirely absent.
- MDHGFN w/o TAEM: In this variant, only the TAEM is removed, while the HSAM is retained. This allows us to investigate the specific impact of the temporal-aware enhanced module on capturing temporal characteristics.
- MDHGFN w/o HSAM: In this variant, only the HSAM is removed, while the TAEM is retained. This variant is used to analyze the role and importance of HSAM in capturing higher-order spatial dependencies.

Table 3 presents the results of the ablation experiments for different model variants on the PEMS03 and PEMS08

Table 2: Performance comparison between different baseline models and MDHGFN on the PEMS03, PEMS04, PEMS07, and PEMS08 datasets, where the best results are highlighted in bold and the second-best results are marked with an underline. Average (Rank) metric column is computed by first calculating the mean performance scores of each model across all evaluation metrics on every dataset, followed by ranking these mean values, where a higher ranking position indicates superior overall performance.

	Method	PEMS03			PEMS04			PEMS07			PEMS08			Average(Rank)		
		MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
2015	FC-LSTM [50]	21.33	35.11	23.33	27.14	18.20	18.20	29.98	45.94	13.20	22.20	34.06	14.20	12	7	11
2017	DCRNN [51]	24.11	38.69	22.67	30.49	45.99	20.72	34.82	51.32	16.23	21.76	32.41	14.19	14	14	13
2018	STGCN [40]	23.48	37.51	21.95	28.80	44.00	19.08	33.80	51.74	14.75	22.49	34.05	14.54	13	13	12
2019	Graph WaveNet [52]	24.73	38.42	24.01	30.55	46.31	23.33	34.62	51.86	15.68	23.01	35.67	15.37	15	15	14
2020	STSGCN [53]	20.72	33.75	19.44	24.21	38.36	15.63	28.10	45.88	11.90	19.56	30.58	12.55	10	12	9
2020	OGCRNN [54]	17.16	29.40	16.48	22.16	35.15	15.72	24.45	39.80	10.22	18.50	29.00	11.74	7	8	7
2021	STGODE [55]	16.50	27.84	16.69	20.84	32.82	13.77	22.59	37.54	10.14	16.81	25.97	10.62	4	4	5
2021	STFGNN [56]	19.37	32.57	18.37	21.55	34.60	14.26	24.77	40.55	10.33	18.83	29.66	11.90	8	10	8
2022	GMSDR [57]	15.92	28.87	14.50	21.63	34.05	13.87	23.78	44.60	10.17	16.59	25.69	10.59	6	6	4
2022	FEDformer [58]	19.74	29.51	20.04	23.57	35.65	50.54	29.42	43.65	22.85	24.22	35.32	17.38	11	11	15
2023	HSTGCNT [59]	19.22	30.65	19.72	23.47	35.53	16.52	24.43	37.77	11.25	20.82	30.87	13.76	9	9	10
2023	AGFCRN [60]	16.89	28.98	17.65	20.51	33.81	13.81	22.69	37.95	9.62	17.35	27.66	11.31	5	5	6
2024	DS-STGCN [14]	16.04	26.40	15.69	19.80	31.81	13.40	22.11	36.62	9.38	15.66	25.65	**10.09**	3	2	2
2024	CDAGF [61]	16.54	27.71	16.28	19.55	32.58	13.06	21.56	37.03	9.11	15.68	25.71	10.13	2	3	3
(**Ours**)	**MDHGFN**	**14.12**	**22.52**	**12.95**	**19.29**	**31.57**	**12.35**	**20.41**	**33.69**	**8.98**	**15.45**	**25.20**	11.50	1	1	1

datasets. The experimental results demonstrate that the complete MDHGFN model outperforms all ablation variants across the RMSE, MAE, and MAPE metrics, validating the critical role of the HSAM and TAEM in spatiotemporal feature modeling. Specifically, removing the HSAM leads to a significant drop in performance, highlighting the importance of higher-order spatial relationships in capturing complex interactions between regions. Similarly, removing the TAEM also results in a decline in performance, reflecting the contribution of temporal feature modeling to prediction accuracy. The model variant with both HSAM and TAEM removed performs the worst, further confirming the synergistic effect of these two modules in enhancing the overall performance of the model.

Table 3: Ablation study results on the PEMS03 and PEMS08 datasets, comparing the performance of different model configurations.

Datasets	Model	RMSE	MAE	MAPE(%)
PEMS03	MDHGFN w/o both HSAM and TAEM	23.48	14.91	13.83
	MDHGFN w/o HSAM	23.07	14.64	13.51
	MDHGFN w/o TAEM	22.68	14.21	13.39
	MDHGFN	**22.52**	**14.12**	**12.95**
PEMS08	MDHGFN w/o both HSAM and TAEM	26.54	16.72	12.93
	MDHGFN w/o HSAM	25.57	15.90	11.89
	MDHGFN w/o TAEM	25.26	15.52	11.82
	MDHGFN	**25.20**	**15.45**	**11.50**

Table 4: Comparison of ablation study results on the PEMS03 and PEMS08 datasets with different combinations of dual hypergraphs.

Datasets	Dual Hypergraph	RMSE	MAE	MAPE(%)
PEMS03	$[G^r]$	22.63	14.23	13.01
	$[G^c]$	22.66	14.30	13.48
	$[G^t]$	22.88	14.53	15.03
	$[G^r, G^c, G^t]$	**22.52**	**14.12**	**12.95**
PEMS08	$[G^r]$	25.49	15.48	11.52
	$[G^c]$	25.36	15.53	12.07
	$[G^t]$	25.52	15.46	11.57
	$[G^r, G^c, G^t]$	**25.20**	**15.45**	**11.50**

Additionally, Table 4 presents the ablation results of different combinations of dual hypergraphs on the PEMS03 and PEMS08 datasets. The experiments focus on the hypergraph convolution within the HSAM, evaluating the performance of using the travel intent dual hypergraph (G^t), the fuctional community dual hypergraph (G^c), and the regional propagation dual hypergraph (G^r) individually, as well as their combined usage. The results demonstrate that the combined usage of the three dual hypergraphs can fully integrate different types of higher-order information, significantly enhancing model performance. This highlights the complementary nature of the three dual hypergraphs in capturing diverse and complex spatial dependencies in traffic networks, thereby improving the overall prediction accuracy.

4.5.3. Effect of Hyperparameters

We conduct hyperparameter experiments on the PEMS08 dataset to evaluate the impact of different configurations of K (the number of KNN neighbors for the functional community dual hypergraph) and Top-k (the number of key nodes selected for the regional propagation dual hypergraph) on model performance. The results are shown in Figure 4.

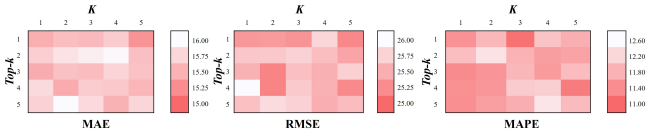


Figure 4: The hyperparameter tuning results on the PEMS08 dataset illustrate the performance of the MDHGFN model under various configurations of K (number of neighbors in KNN) and Top-k (number of key nodes selected for feature filtering). In this visualization, darker colors indicate better performance.

The hyperparameter settings significantly influence the construction of the multi-scale hypergraph's topological structure, thereby affecting the model's ability to capture features across multiple scales, including individual travel intentions at the micro level, community commuting patterns at the meso level, and regional flow propagation at

the macro level. Experimental results demonstrate that fine-tuning these hyperparameters effectively balances feature sufficiency and redundancy, fully leveraging the complementary advantages of the functional community dual hypergraph and the regional propagation dual hypergraph. This optimization process not only enhances the model's capability to extract high-order spatiotemporal relationships but also validates the fundamental value of the proposed dual hypergraph construction method in accurately modeling complex spatiotemporal dependencies in transportation networks. Through systematic parameter optimization, we identified the optimal configuration (K=2 and Top-K=4), which achieves the best trade-off between model complexity and prediction accuracy.

5. Conclusions

This research proposes a Multiscale Dual Hypergraph Fusion Spatiotemporal Network (MDHGFN) with multiscale high-order spatial awareness for complex urban traffic flow prediction. The main contributions of this research are threefold. First, to address the limitations of existing models in characterizing the ubiquitous multiscale high-order spatial features of traffic networks, we introduce a multiscale dual hypergraph construction method, which systematically models high-order spatial characteristics of urban traffic at three scales: microscopic individual travel intent, mesoscopic community commuting patterns, and macroscopic regional flow propagation. Second, design a comprehensive prediction framework integrating both spatial-aware and temporal-aware blocks. The spatial-aware block employs hypergraph convolution and dynamic graph convolution to extract dual hypergraph features progressively, incorporates a gating mechanism to fuse high-order local spatial information from multiscale dual hypergraphs, and further utilizes spatial multi-head attention to capture global spatial dependencies. The temporal-aware block effectively extracts both short-term fluctuations and long-term trends in traffic flows through dynamic temporal dilated causal convolution and temporal multi-head attention. Finally, experimental results on four real-world traffic flow datasets demonstrate that the proposed model significantly outperforms 14 state-of-the-art baseline models across multiple evaluation metrics, exhibiting superior prediction performance and generalization capability. This research provides a novel and effective solution for high-precision spatiotemporal prediction in complex traffic scenarios.

CRediT authorship contribution statement

Jiajun Xian: Conceptualization, Methodology, Project administration, Software, Validation, Writing – original draft. **Yixuan Ye**: Methodology, Validation. **Wei Zhang**: Methodology, Validation. **Zishen Chen**: Methodology, Validation. **Jiaqiang Huang**: Methodology, Validation. **Zichun Lin**: Methodology, Validation. **Can Zou**: Methodology, Validation. **Hao LIU**:Methodology, Validation. **Dan Yang**: Methodology, Validation, Writing – original draft. **Nan Meng**: Methodology, Validation. **Ming Liu**: Methodology, Validation, Writing – review & editing. **Teng Zhou**: Supervision, Writing – review & editing.

Data availability

The dataset and source code generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Conflicts of interest

The authors declare that they have no conflict of interest.

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